

$$\Rightarrow \frac{2}{d} = \frac{1}{c} + \frac{1}{e} \quad \dots(iii)$$

from (i) and (ii)

$$2c^2 = (a + c)d$$

$$\Rightarrow \frac{2}{d} = \frac{a+c}{c^2}$$

$$\Rightarrow \frac{1}{c} + \frac{1}{e} = \frac{a}{c^2} + \frac{1}{c} \quad \text{from (iii)}$$

$$\Rightarrow \frac{1}{e} = \frac{a}{c^2}$$

$$\Rightarrow c^2 = ae$$

So, a, c, e are in GP.

27. Option (b) is correct.

$$\begin{aligned} \log_4(x-1) &= \log_2(x-3) \\ \Rightarrow \frac{1}{2} \log_2(x-1) &= \log_2(x-3) \\ \Rightarrow \log_2(x-1) &= \log_2(x-3)^2 \\ \Rightarrow x-1 &= x^2 - 6x + 9 \\ \Rightarrow x^2 - 7x + 10 &= 0 \\ \Rightarrow x^2 - 5x - 2x + 10 &= 0 \\ \Rightarrow (x-5)(x-2) &= 0 \\ x &= 2, 5 \text{ But } x-3 > 0, x > 3 \\ \therefore x &= 5 \text{ only solution.} \end{aligned}$$

28. Option (d) is correct.

$$\begin{aligned} \log_x \left(\frac{x}{y} \right) + \log_y \left(\frac{y}{x} \right) &= k \\ \Rightarrow \log_x x - \log_x y + \log_y y - \log_y x &= k \\ \Rightarrow 1 - \log_x y + 1 - \frac{1}{\log_x y} &= k \\ \text{Let } \log_x y &= t \\ \therefore 2 - t - \frac{1}{t} &= k \\ \Rightarrow 2t - t^2 - 1 &= kt \\ \Rightarrow t^2 + (k-2)t + 1 &= 0 \\ \text{For solution } D &= b^2 - 4ac \geq 0 \\ (k-2)^2 - 4 \geq 0 &\Rightarrow (k-2)^2 \geq 4 \\ k-2 \geq 2 &\quad k-2 \leq -2 \\ k \geq 4 &\quad \text{or} \quad k \leq 0 \\ \therefore k &\neq 1 \end{aligned}$$

29. Option (c) is correct.

$$|A| = \begin{vmatrix} \sin 2\theta & -\cos 2\theta & 0 \\ \cos 2\theta & \sin 2\theta & 0 \\ 0 & 0 & 1 \end{vmatrix} = 1$$

We know that if $|A| = \pm 1$ then A is orthogonal matrix.

$$\text{So, } A^{-1} = A^T \text{ and } A^{-1} = \text{adj } A$$

30. Option (a) is correct.

$$\begin{aligned} (1-x^2)^{20} \left[-\left(\frac{1}{x^2} + x^2 - 2 \right) \right]^{-5} \\ = -(1-x^2)^{20} \left(x - \frac{1}{x} \right)^{-10} \\ = -(1-x^2)^{20} (x^2 - 1)^{-10} x^{10} \\ = -(1-x^2)^{10} x^{10} \\ T_{r+1} = {}^{10}C_r (1)^{10-r} (-x^2)^{r-10} x^{10} \\ = (-1)^{11-r} {}^{10}C_r x^{30-2r} \\ \therefore 30-2r = 10 \\ \Rightarrow r = 10 \\ \therefore \text{Coefficient of } x^{10} = (-1)^1 {}^{10}C_{10} \\ = -1 \end{aligned}$$

31. Option (b) is correct.

$$\begin{aligned} T_4 &= T_{3+1} \\ &= {}^nC_3 (mx)^{n-3} \left(\frac{1}{x} \right)^3 = \frac{5}{2} \\ \frac{5}{2} x^0 &= {}^nC_3 (m)^{n-3} x^{n-6} \end{aligned}$$

On comparing the power of x

$$\begin{aligned} \therefore n-6 &= 0 \\ \Rightarrow n &= 6 \\ \text{and } {}^6C_3 m^3 &= \frac{5}{2} \\ \Rightarrow \frac{6 \cdot 5 \cdot 4}{3 \cdot 2} m^3 &= \frac{5}{2} \\ \Rightarrow m^3 &= \frac{1}{8} \\ \Rightarrow m &= \frac{1}{2} \\ \therefore mn &= 6 \times \frac{1}{2} = 3 \end{aligned}$$

32. Option (d) is correct.

Given that a, b, c are in GP.

$$\begin{aligned} \therefore b^2 &= ac \\ \text{Now, for } ax^2 + bx + c &= 0 \\ D &= b^2 - 4ac \\ &= ac - 4ac = -3ac < 0 \end{aligned}$$

$\therefore$ Roots are imaginary

Let $a = 2, b = 4, c = 8$

$$\begin{aligned} \therefore x^2 + 2x + 4 &= 0 \\ \Rightarrow x &= \frac{-2 \pm \sqrt{4-16}}{2} \\ &= -1 \pm \sqrt{3} \end{aligned}$$